FactBench: A Dynamic Benchmark for In-the-Wild Language Model Factuality Evaluation

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https://huggingface.co/spaces/launch/factbench

Abstract

The rapid adoption of language models (LMs) across diverse applications has raised concerns about their factuality, i.e., their consistency with real-world facts. We first present VERIFY (**V**erification and **E**vidence **R**etr**I**eval for **F**actualit**Y** evaluation), a pipeline to evaluate LMs' factuality in real-world user interactions. VERIFY considers the verifiability of LM-generated content and categorizes content units as `supported`, `unsupported`, or `undecidable` based on Web-retrieved evidence. Importantly, factuality judgment by VERIFY correlates better with human evaluations than existing methods. Using VERIFY, we identify "hallucination prompts" across diverse topics, i.e., those eliciting the highest rates of incorrect (`unsupported`) and inconclusive (`undecidable`) LM responses. These prompts form FACTBENCH, a dataset of 1K prompts across 150 fine-grained topics. Our dataset captures emerging factuality challenges in real-world LM interactions and can be regularly updated with new prompts. We benchmark widely-used LMs from GPT, Gemini, and Llama families on FACTBENCH, yielding the following key findings: **(i)** Proprietary models exhibit better factuality, with decreased performance from `Easy` to `Hard` hallucination prompts. **(ii)** Llama3.1-405B-Instruct shows comparable or lower factual precision than Llama3.1-70B-Instruct across all evaluation methods due to its higher subjectivity that leads to more content labeled as `undecidable`. **(iii)** Gemini1.5-Pro shows a significantly higher refusal rate, with over-refusal in 25% of cases.

1 Introduction

Despite ongoing efforts to enhance their factuality, language models (LMs) continue to generate false or irrelevant content, often known as *hallucination* (Huang et al., 2023; Liu et al., 2023). The widespread use of LMs and the evolving nature of information demand a dynamic factuality evaluation benchmark to identify the challenges LMs face in real-world applications. Current long-form factuality evaluation benchmarks (Min et al., 2023; Wei et al., 2024b; Malaviya et al., 2024) are limited by being static and have a narrow coverage of usage scenarios. The static design makes these benchmarks susceptible to data leakage (Magar and Schwartz, 2022), rendering them unsuitable for capturing factuality challenges in real-world LM usages. Moreover, existing benchmarks often focus on a sparse subset of tasks. For instance, data used in developing FactScore (Min et al., 2023) primarily addresses biographical queries, while ExpertQA (Malaviya et al., 2024) recruits human experts to curate domain-specific questions. Other benchmarks (Chen et al., 2023; Wei et al., 2024b) cover queries that are either LM-generated or human-curated, limiting their applicability to in-the-wild scenarios.

In this work, we make two primary contributions: **(i)** a dynamic and diverse factuality evaluation benchmark curated from real-world LM usage and **(ii)** a factuality evaluation framework that measures the appropriateness of in-the-wild prompts for inclusion in our benchmark by estimating how frequent strong LMs generate incorrect and inconclusive responses. Concretely, we introduce **FACTBENCH**, an updatable benchmark grounded in the real-world usage of LMs. FACTBENCH comprises 1,000 diverse information-seeking prompts across 150 topics (examples in Figure 2), that can cause LMs to produce false and inconclusive responses. Using clustering methods, we begin by identifying 382 unique tasks within the LMSYS-Chat-1M dataset (Zheng et al., 2024). Prompts within each task cluster are then labeled for verifiability, indicating whether the prompt's response can be verified using Google Search results. We then assess the usefulness of prompts by considering factors such as clarity, interest, and relevance to a broad audience. Finally, verifiable prompts that meet a

specified usefulness threshold become candidates for inclusion in FACTBENCH.

We further define the "hallucination prompts" as the ones that can elicit incorrect or inconclusive content from a selected group of strong LMs (e.g., GPT-4 and Claude-2). To identify such prompts, we design **VERIFY**, a **V**erification and **E**vidence **R**etr**I**eval for **F**actualit**Y** evaluation pipeline to detect nonfactual content in LM responses. VERIFY first extracts content units from model responses and identifies their type (e.g., facts, instructions, disclaimers, etc.). It then evaluates only the verifiable units against Web-based evidence using an interactive query generation and evidence retrieval technique. Finally, VERIFY categorizes units as `supported`, `unsupported`, or `undecidable` based on the evidence.

LMs should ideally generate factual outputs that can be verified against reliable sources. To quantify the degree of hallucination in model responses, we propose a hallucination score that penalizes both incorrect claims (`unsupported`) and claims that cannot be verified due to ambiguity, partial correctness, or missing context (`undecidable`). This score measures the appropriateness of the corresponding user prompts for our final benchmark. We categorize prompts into three tiers (`Hard`, `Moderate`, and `Easy`) based on the responding models' strength and select those with the highest hallucination scores within each tier. This results in a final benchmark of 1,000 prompts after manual inspection.

To study trends in factuality evaluation methods and compare their effectiveness, we focus on four representative LMs: two proprietary models, i.e., GPT4-o (OpenAI, 2024a) and Gemini1.5-Pro (Team et al., 2024), and two open-weight models from Llama3.1 family, i.e., Llama3.1-70B-Instruct and Llama3.1-405B-Instruct (Meta, 2024) (full benchmarking results across seven models are in Appendix 10.10). The results show that LM performance significantly increased across tiers, aligning with our curation strategy. To compare the effectiveness of different factuality evaluation methods, we use VERIFY units as a common basis and feed them into factuality evaluation baselines for verification. Our results reveal that VERIFY achieves the highest correlation with human judgments compared to state-of-the-art methods. This finding underscores the effectiveness of our approach in benchmark creation and factuality assessment.

In summary, our contributions are as follows:

- We introduce FACTBENCH, a new benchmark grounded in the real-world usage of LMs. FACTBENCH is designed to be updatable by continuously incorporating newly collected hallucination prompts that cause LMs to generate incorrect content. This dynamic approach ensures that the benchmark remains relevant, addressing the evolving challenges in factual generation.
- We design VERIFY, a factuality evaluation pipeline that considers the verifiability of generated content and categorizes units into `supported`, `unsupported`, or `undecidable` according to retrieval results. VERIFY addresses the limitations of prior work that makes binary decisions on supportedness and achieves the highest average correlation with human evaluations.
- We release factuality annotations by humans on 4,467 content units, with each unit independently judged by two annotators. Each annotator evaluates the independence of extracted units and their factuality using Google Search. This human-annotated data provides quantifiable evaluation resources for assessing future factuality evaluation techniques.

2 Related Work

2.1 Factuality Evaluation Benchmarks

The widespread adoption of LMs, coupled with their tendency to hallucinate, demands new benchmarks that can effectively identify their factual weaknesses across diverse scenarios. Prior factuality evaluation benchmarks mainly focus on short-form and human-curated question-answering (QA) tasks. For instance, TruthfulQA (Lin et al., 2022), HaluEval (Li et al., 2023), and FELM (Chen et al., 2023) mostly focus on short-form knowledge-based QA with questions with human-selected topics, despite LMs typically engaging in long-form conversations. The data used in developing FactScore (Min et al., 2023), while long-form, is limited to a single, relatively easy task of biographical QA. LongFact (Wei et al., 2024b) expands to 38 human-selected topics, but the prompts are LM-generated rather than user-driven. FactCheck-Bench (Wang et al., 2024a) collects ChatGPT hallucinations from Twitter, but its scope is narrow (94 prompts) and focuses on a specific and rather obsolete model. Moreover,

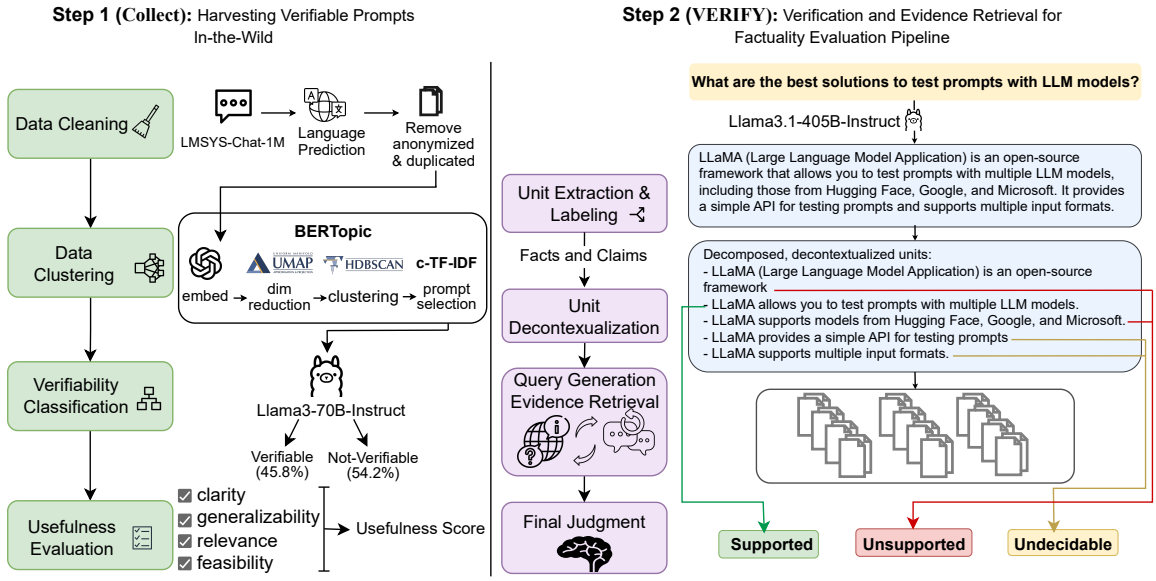


Figure 1: This figure outlines the two-step process we use to evaluate LM responses. Step 1 (left) involves cleaning, clustering, and evaluating prompts for verifiability and usefulness. Step 2 (right) decomposes responses into units, retrieves external evidence, and generates factuality labels (`supported`, `unsupported`, `undecidable`) with a hallucination score to flag inaccuracies. This process involves the collection and appropriateness assessment of hallucination prompts.

all these datasets are static and prone to the data contamination issues (Magar and Schwartz, 2022). We fill these gaps by offering a benchmark that systematically mines hallucination prompts from in-the-wild user-model chat logs in LMSYS-chat-1M (Zheng et al., 2024), spanning diverse topics. FACTBENCH is designed to be regularly updated with new real-world prompts, ensuring relevance to evolving LM capabilities and use cases rather than being limited to fixed timeframes.

2.2 Factuality Evaluation Methods and Verifiability

The challenge of distinguishing verifiable from non-verifiable claims is central to fact-checking. AFaCTA (Ni et al., 2024) stresses that claims are verifiable when they provide sufficient specificity for evidence retrieval. The subjective nature of check-worthiness, shaped by political and social contexts (Konstantinovskiy et al., 2020; Nakov et al., 2022), complicates this, particularly in LM-generated content where fact-opinion lines blur (Vosoughi et al., 2018). To address this, VERIFY introduces an `undecidable` label for claims with ambiguous factuality, accommodating both objective and context-dependent claims.

Long-form content evaluation presents unique challenges due to its complexity and the numerous claims it typically contains. To address these challenges, SAFE (Wei et al., 2024b) and FactScore (Wang et al., 2024b) decompose content into individual facts for granular verification. Our method, VERIFY, builds upon this approach by decomposing LM-generated content into units and distinguishing between verifiable and non-verifiable elements that appear in user-model interactions. While VeriScore (Song et al., 2024) similarly recognizes that not all LM-generated content is verifiable, its single-step approach to extracting and decontextualizing *verifiable claims* may limit its effectiveness in complex scenarios. Moreover, VeriScore's fixed-size context window could fail to capture long-range dependencies in real-world responses, potentially missing crucial context for accurate fact-checking. We address these limitations through a multi-step process that leverages the complete LM response to carefully identify independent verifiable content units. Our pipeline then evaluates these verifiable units by classifying them as `supported` or `unsupported` only when confident evidence is found, labeling them `undecidable` otherwise. This approach introduces a more robust framework for evaluating the factual precision of LM-generated content. In contrast, Factcheck-GPT (Wang et al., 2024a) is less reliable for factual evaluation due to its coarser content extraction and dependence on the model's parametric knowledge when external evidence is unavailable.

3 Harvesting Verifiable Prompts From LM Interactions

Our current understanding of LM performance on verifiable tasks is limited, and existing factuality evaluation benchmarks cover only a narrow range of verifiable use cases. To address this gap, we collect English prompts from the first turn of conversations in the LMSYS-Chat-1M dataset (Zheng et al., 2024), which is a large-scale, in-the-wild LM conversations dataset. Our objective is to identify a set of verifiable and useful prompts across diverse topics through a multi-step process described below. Figure 1 (left) outlines our collection process.

- **Data Clustering**: After cleaning the data (see details in Appendix 10.1), we get 294,333 distinct prompts and cluster them into various topics. We use BERTopic (Grootendorst, 2022), a dynamic topic-modeling pipeline that (1) embeds prompts using OpenAIs text-embedding-3-small model (OpenAI, 2024b), (2) applies UMAP (McInnes et al., 2020) for dimensionality reduction, and (3) employs HDBSCAN (Malzer and Baum, 2020), a hierarchical density-based clustering algorithm. This algorithm is particularly effective in discovering clusters when their number and density distribution are unknown while requiring minimal parameter tuning (see Appendix 10.3 for details). HDBSCAN identifies 142,702 (48.5%) of the prompts as outliers, which we exclude to remove overly-specific prompts referring to niche user requests. Finally, we use a class-based TF-IDF method to select the top 100 most representative prompts from each cluster and summarize them into concise topics (up to 10 words) using GPT-4 Turbo (OpenAI, 2024c). This results in 382 clusters with verbalized topic examples demonstrated in Appendix Figure 9.

- **Verifiability Classification**: We focus on prompts that elicit responses with varying degrees of verifiable information. To identify these, we employ Llama3-70B-Instruct (AI@Meta, 2024) to distinguish between verifiable and non-verifiable prompts (see Appendix 10.15.2 for classification prompt and Figure 9 for proportions of verifiable prompts across clusters). Overall, verifiable prompts constitute 45.8% of total prompts from the previous step.

- **Usefulness Evaluation.** The remaining collection contains around 70K prompts, too large for manual or automated fact-checking to identify "hallucination prompts" as described in Section 4. Randomly selecting a subset for evaluation would be suboptimal, as it may include unclear or over-specific requests. Therefore, we propose a set of criteria to identify *useful* prompts. A useful prompt needs to be (**i**) clear and understandable, (**ii**) generalizable to various users or scenarios, (**iii**) potentially interesting or valuable to a broader audience, and (**iv**) is within the capabilities of LMs (e.g., excludes real-time data). To reduce single-model bias, we use two language models, GPT-4-Turbo and Llama3-70B-Instruct, as annotators. Each model scores prompts on a scale of 1 (low) to 5 (high) for each criterion. The final usefulness score for each prompt is calculated as the average score across all criteria, summing the score from two models (see Appendix 10.4 for detailed score calculation). The usefulness scores are then used to select an initial set of prompts, as described in Section 5.

4 VERIFY: Verification and Evidence Retrieval for Factuality Evaluation

In this section, we present VERIFY, an automatic factuality evaluation framework that measures prompt *appropriateness* for inclusion in FACTBENCH. We quantify a prompt's appropriateness through its ability to elicit hallucinations in the responding LM. Section 4.1 first establishes criteria for determining the verifiability of statements. We then present our framework's two components: (1) an evaluation pipeline that automatically labels responses for factual accuracy (Sections 4.2–4.5), and (2) a hallucination score that integrates these labels into a final metric (Section 4.6).

4.1 Verifiability-driven Factual Evaluation

A statement is verifiable if it provides sufficient information to guide fact-checkers in verification (Ni et al., 2024). We classify *verifiable statements* into two categories:

Context-independent Statements: These are objective assertions that can be directly verified against knowledge sources. For example, "RTX 3060 has a memory bandwidth of 360 Gbps."

Context-dependent Statements: These statements require additional information for verifica-

tion. For instance, to verify the statement "The difference in memory bandwidth between the RTX 3060 and RTX 3060 Ti is relatively small", one needs knowledge of both GPUs' bandwidths and an understanding of what qualifies as *relatively small* in this context.

In natural user-LM settings, model responses may include a mixture of verifiable and non-verifiable statements. By focusing on verifiable statements, we enable a more precise factuality assessment of responses, as described next.

4.2 Unit Extraction and Labeling

User requests span a wide range of domains (examples provided in Figure 2), and model responses contain a variety of content types. To evaluate verifiable statements, we first decompose the model response into independent content units. A content unit can represent a `Fact`, a `Claim`, an `Instruction`, a `Disclaimer`, a `Question`, or other relevant content type that appears during user-model interaction. A taxonomy of content types is detailed in Appendix 10.15.4 as part of the prompt description. Many content units, such as questions and disclaimers, are not verifiable as they convey conversation context or model capabilities rather than factual information.

We label each unit by its type to identify those suitable for verification. We use the Llama3-70B-Instruct model, which serves as the backbone LM for all tasks in this pipeline, to extract and label content units. A carefully crafted prompt with examples (see Appendix 10.15.4) guides the model in this task. We classify objective statements as `Fact` and potentially subjective, context-dependent statements as `Claim`. Only units labeled as `Fact` or `Claim` are passed to the next step.

4.3 Unit Decontextualization

Gunjal and Durrett (2024) highlights the importance of "molecular units"—units that contain sufficient information to be uniquely identifiable in factuality assessment. Inspired by that, we implement a unit decontextualization component in our pipeline to minimally revise verifiable units and make them self-contained. The prompt is provided in Appendix 10.15.5.

4.4 Query Generation and Evidence Retrieval

In order to verify the self-contained units, we need to retrieve relevant evidence from knowledge sources. We utilize SerperAPI[1] for Google Search and Web-evidence retrieval.

To enhance search quality and ensure the retrieval of relevant evidence that best assists in verification, we implement an interactive query refinement technique. The query generator operates iteratively within an interactive feedback loop. It first produces a query for the target unit, which is then used in Google Search to return relevant snippets. In subsequent rounds, the query generator evaluates the relevance of the retrieved snippets for verifying the target unit and refines the query accordingly. Through empirical analysis, we found that five rounds consistently produce high-quality, relevant queries. The final set of queries and associated search results are then passed to the next step for final judgment. The prompt is provided in Appendix 10.15.6.

4.5 Final Answer Generation

In this step, the model (Llama3-70B-Instruct) makes final decisions on extracted units' factual accuracy by evaluating retrieved evidence through Chain-of-Thought prompting (Wei et al., 2024a). For each unit, the model: **(i)** summarizes relevant knowledge points, **(ii)** assesses their relationship to the unit, and **(iii)** classifies the evidence as either `supported`, `unsupported`, or `undecidable`. The prompt template is provided in Appendix 10.15.7. This process produces annotation labels for all verifiable units in the original model response. An overview of the evaluation pipeline is illustrated in the right part of Figure 1. While we instantiate VERIFY with Llama3-70B-Instruct in this work, the framework facilitates affordable factuality checking by leveraging open-weight LMs.

4.6 Hallucination Score

After annotating the individual content units, we propose a hallucination metric to quantify the incorrect and inconclusive contents within a model's response. Let US represent the set of `unsupported` units, UD the set of `undecidable` units, and V the total set of extracted verifiable units (`Claims` and `Facts`). The final *Hallucination Score* is computed as follows:

$$H(R) = \frac{|US| + \alpha|UD|}{\sqrt{|V|}} \quad (1)$$

Here, $\alpha \in (0, 1)$ determines the relative weight of `undecidable` units compared to `unsupported`

[1] https://serper.dev/

Benchmark	In-the-Wild	Dynamic	# Prompts
FELM (Chen et al., 2023)	✗	✗	847
ExpertQA (Malaviya et al., 2024)	✗	✗	484
FactScore (Min et al., 2023)	✗	✗	500
LongFact (Wei et al., 2024b)	✗	✗	2280
FactCheckBench (Wang et al., 2024a)	mixed	✗	94
FACTBENCH	✓	✓	1000

Common Factual Requests

8.9% Travel itineraries
6.5% Medical questions
4.9% Recipe requests
2.9% LM apps (i.e. in education)
2.1% GPU recommendations
2.1% Game comparisons
1.9% Music recommendations
1.9% Relativity exploration
1.8% Solar system inquiries
1.7% Transformers exploration

Figure 2: Statistics of different factuality benchmarks. FACTBENCH is the first dynamic and in-the-wild factuality evaluation benchmark with diverse topic coverage.

ones, reflecting cases such as: (1) potentially accurate information that lacks sufficient context or originates from outdated knowledge in the training data, (2) information that cannot be verified through current web searches, and (3) plausible but unverifiable combinations of existing facts. Based on manual analysis (details in Appendix 10.5), we set $\alpha = 0.5$. The denominator $\sqrt{|V|}$ grows sublinearly with the number of verifiable units to ensure that the score remains sensitive to errors even in longer responses. This design choice balances the score across responses of varying lengths while preventing large sets of verifiable units from masking error significance.

5 FactBench Dataset

The hallucination score helps us assess prompt appropriateness for our final dataset. To ensure balanced representation across model capabilities, we categorize prompts into three tiers: `Hard`, `Moderate`, and `Easy`, based on the overall performance of responding models[2]. The `Hard` tier consists of prompts queried to the strongest models in LMSYS-Chat-1M (e.g., GPT-4, Claude-2), preventing weaker models' higher hallucination rates from dominating our dataset. The full list of models for each tier is provided in Appendix Table 5.

For prompt selection, we apply tier-specific usefulness thresholds: 4 or higher for `Hard`, 4.5 or higher for `Moderate`, and exactly 5 for `Easy`. This approach reflects our assumption that responses from stronger models better indicate prompt appropriateness, hence allowing a lower threshold for the `Hard` tier. From the 70K prompts collected in Section 3, this filtering yields 4.2K prompts distributed as 53% `Hard`, 34% `Moderate`, and 13% `Easy`.

We then evaluate these prompts using the hallucination score (Equation 1) of their corresponding LM responses. We select 1K prompts with the highest scores while maintaining the original tier distribution (532 `Hard`, 332 `Moderate`, 136 `Easy`), then iteratively filter out-of-scope prompts through manual inspection as detailed in Appendix 10.2. Figure 2 (left) compares our benchmark statistics with other long-form factuality evaluation benchmarks. Our work introduces the first real-world factuality evaluation benchmark comprising hallucination prompts across diverse topics. We intend to regularly update FACTBENCH as more prompts are added to the LMSYS-chat-1M data (see Appendix 10.6 for details of our updating process).

6 Experimental Setup

Language Models[3]: We benchmark LMs against FACTBENCH to evaluate their performance on this dataset using different factuality evaluation methods. We evaluate recent powerful models (available via APIs) from proprietary and open-source categories. From the proprietary models, we benchmark GPT-4o (omni) (OpenAI, 2024a) and Gemini1.5-Pro (Team et al., 2024). For open-source models, we evaluate Llama3.1-70B-Instruct and Llama3.1-405B-Instruct (Meta, 2024).

Baselines: For comparison, we consider three reference-dependent factuality evaluation techniques: FactScore (Min et al., 2023), Search-Augmented Factuality Evaluator (SAFE) (Wei

[2]Model performance rankings are based on the Chatbot Arena Leaderboard, which ranks LMs through pairwise human comparisons

[3]Additional LM benchmarking results on FACTBENCH can be found in Appendix 10.10. This list will be continuously expanded as new models become available.

et al., 2024b), and Factcheck-GPT (Wang et al., 2024a). Appendix 10.7 details these methods and their experimental setup.

7 Experimental Results and Further Analyses

In this section, we measure LMs' factuality on FACTBENCH using different factuality evaluation methods (Section 7.1). These methods extract content units at different granularities, and thus, we establish a unified evaluation framework with human labeling (Section 7.2) for fair comparison.

7.1 Factuality Improves with Proprietary LMs and on Easier Prompts

To compare model performance on FACTBENCH, we use the factual precision metric proposed by Min et al. (2023). This metric quantifies an LM's factuality by calculating the proportion of `supported` units among all extracted units in response, averaged across all responses (detailed in Appendix 10.8). Table 1 compares the factual precision of LMs on FACTBENCH as measured by different evaluation methods. Despite all baselines verifying responses at their finest granularity (standalone units), we observe varying factual precision ranges across different pipelines. We observe that VERIFY consistently maintains the same ranking of models across all three tiers, unlike other evaluation methods. VERIFY ranks **GPT4-o as having the highest factual precision across all tiers, followed by Gemini1.5-Pro and the two open-source Llama3.1 models**.

Surprisingly, according to VERIFY, Llama3.1-405B-Instruct performs comparably or worse than its 70B variant. Further investigation (Figure 3) reveals that Llama3.1-405B-Instruct responses contain a lower proportion of units labeled as `unsupported` but the highest proportion of `undecidable` units across all LMs. This primarily stems from stronger subjectivity in Llama3.1-405B-Instruct, with more frequent use of subjective adjectives such as "solid," "exclusive," and "well-known." The reasoning logic enforced in our pipeline typically classifies these subjective units as `undecidable`, thereby reducing the factual precision. Detailed analysis and examples are provided in Appendix 10.14.

Another significant observation is **the consistent decline in factuality precision across LMs from the `Easy` to `Hard` tiers**, as shown by all evaluation methods. This validates with our tiered benchmark design, where prompts are categorized based on the responding model's strength. `Easy` prompts are less likely to trigger hallucinations in strong models, as their appropriateness is measured based on weaker LM hallucinations.

7.2 Factuality Evaluation by VERIFY Strongly Correlates with Human

The factuality of a model, measured by a factuality evaluation method, depends on the granularity of the extracted units and the method's verification capabilities.

FactScore extracts units with finer granularity than VERIFY due to its focus on biographical texts, where units are typically discrete and easily separable. This excessive segmentation removes the necessary context for verification. On the other hand, Factcheck-GPT's claim-level decomposition (finest-level) often results in sentence-level containing multiple claims. To establish a unified evaluation framework for these methods, we collected model responses to 40 randomly sampled FACTBENCH prompts, each from a different topic. We then applied our unit extraction (Section 4.2) and decontextualization approach (Section 4.3) to decompose generated LM responses into *self-contained* and *verifiable* units. This method was chosen for its ability to handle user-model conversations (with careful instructions and in-the-wild demonstrations), extract moderately granular units, and filter them based on verifiability.

Three fluent English speakers are hired to annotate a total of 160 LM responses for four models on the same set of 40 prompts. VERIFY breaks LM responses into 4,467 units, with each response annotated by two annotators. Each annotator evaluated the independence and factuality of each unit. A unit is considered *Independent* if it is *verifiable* and *self-contained*. A *Dependent* unit, on the other hand, is either an *unverifiable* piece of information, e.g., "I can provide you with some examples." or *under-specified* content. For example, "She won the best actress award" is underspecified as it contains a pronoun, and it is unclear what specific award is referenced. Overall, 82.6% units are considered *Independent* by both annotators with an inter-annotator agreement of 0.53 in terms of Cohen's Kappa score. Additionally, annotators evaluate unit factuality using two labels: *Factual* for statements with supporting Web evidence, and *Other* when evidence either refutes the statement

FactBench	Model	FactScore	SAFE	Factcheck-GPT	VERIFY
Tier 1: Hard	GPT4-o	57.09	**67.42**	**87.43**	**75.65**
	Gemini1.5-Pro	55.51	64.74	84.08	73.78
	Llama3.1-70B*	**57.27**	65.82	82.82	70.07
	Llama3.1-405B*	56.81	64.81	83.77	68.59
Tier 2: Moderate	GPT4-o	**59.27**	**70.25**	**90.85**	**80.72**
	Gemini1.5-Pro	56.59	67.01	87.99	78.02
	Llama3.1-70B*	58.44	68.74	86.38	75.76
	Llama3.1-405B*	57.92	67.82	86.60	75.05
Tier 3: Easy	GPT4-o	**73.01**	79.27	**94.57**	**91.63**
	Gemini1.5-Pro	68.66	77.64	91.97	89.86
	Llama3.1-70B*	73.05	**80.01**	94.08	89.30
	Llama3.1-405B*	73.34	78.87	93.66	86.57

Table 1: Results of VERIFY and baselines across three tiers of FACTBENCH and 4 widely-used LMs (*Instruct version) using the factual precision score (Equation 3), as in prior work (Min et al., 2023; Wei et al., 2024b). For each evaluation method and within each tier, the best and second-best factuality scores are highlighted in blue and green, respectively. Proprietary models are more accurate, with factuality declining from Easy to Hard prompts.

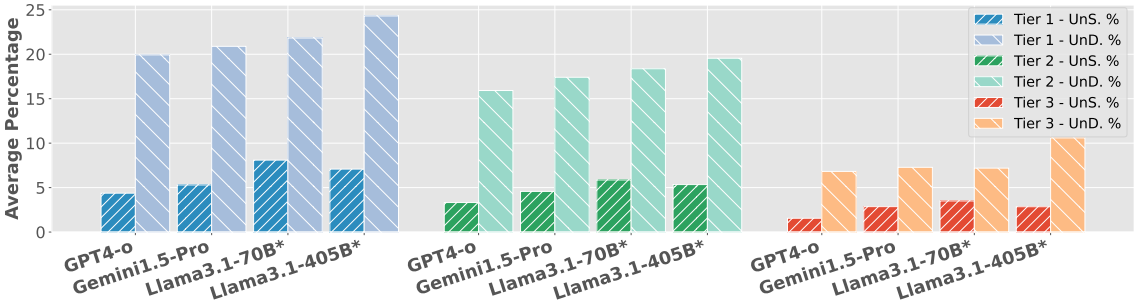


Figure 3: Average percentage of unsupported (UnS) and undecidable (UnD) labels in different LMs (*Instruct version) evaluated by VERIFY. Responses from Llama3.1-405B-Instruct contain the highest proportion of undecidable units across all LMs.

or its factuality cannot be determined. Annotators achieve 85.9% agreement on factuality labels (Cohen's Kappa of 0.57). Binary factuality labels are compatible with VERIFY and other baselines. A unit is labeled *Independent* if both annotators agree and *Dependent* otherwise. Factuality is decided in the same manner.

7.2.1 Accuracy Overlooks Decision Quality

We fed only *Independent* units, as agreed by both annotators, to the factuality evaluation methods for verification. Figure 6 compares the accuracy of different methods against human labels. While VERIFY demonstrates on-par or superior performance on LLaMA3.1 models compared to other baselines (detailed accuracy results in Appendix Figure 7), it is generally outperformed by Factcheck-GPT.

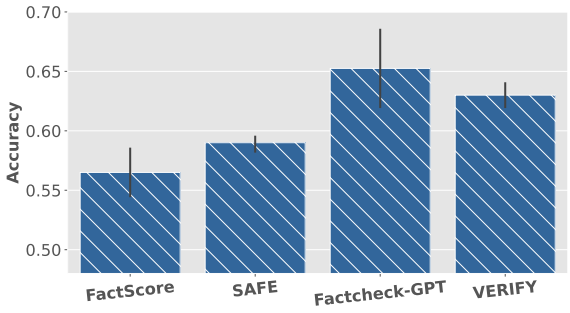


Figure 6: Average accuracy of factuality evaluation methods compared to human annotations across LMs.

Further analysis reveals that Factcheck-GPT achieves the highest average precision in predicting factual units (84.6% compared to VERIFY's 75.1%) across models. This performance gap is attributed to the distinct design choices of the two methods. Factcheck-GPT leverages its internal

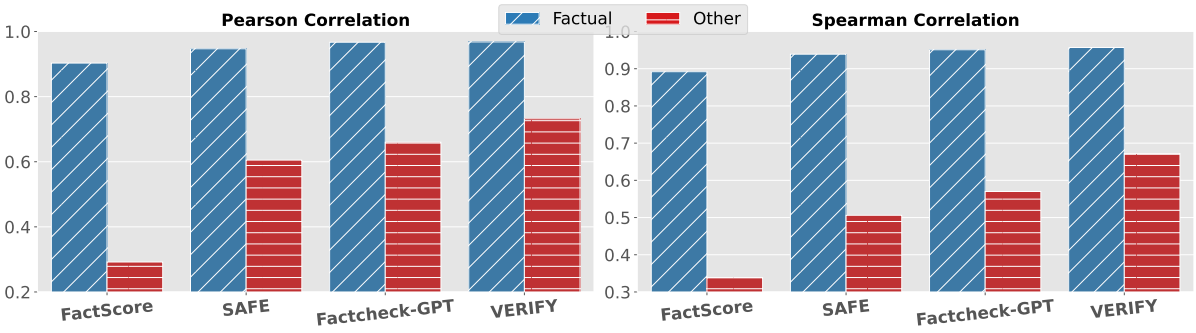


Figure 4: Response-level correlation between factuality evaluation methods and human annotations of 40 prompts across 4 LMs (averaged via z-score). **F** refers to `Factual` labels, and **O** refers to `Other`. VERIFY achieves the highest correlation with human judgments compared to baselines.

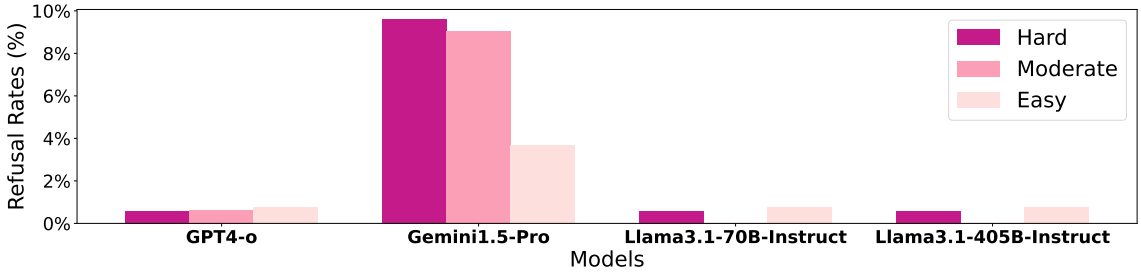


Figure 5: Refusal rate of different LMs across `Hard`, `Moderate`, and `Easy` tiers of FACTBENCH. Gemini1.5-Pro shows a significantly higher refusal rate than other LMs.

knowledge when no external evidence is found, whereas VERIFY takes a conservative approach by labeling such cases as `undecidable`. While this evidence-based strategy may yield lower factuality scores, it ensures more trustworthy evaluations.

To better capture the human-method decision alignment, we calculate correlation following previous work (Wei et al., 2024b; Min et al., 2023). As shown in Figure 4, **VERIFY achieves the highest correlation with human labels compared to other evaluation methods**. Notably, VERIFY achieves a significantly higher correlation with human annotation in the *Other* category. This reflects our method's nuanced handling of `undecidable` cases and demonstrates how VERIFY's judgments mirror human reasoning in scenarios where information cannot be definitively assessed. VERIFY achieves comparable performance to FactCheck-GPT in the *Factual* category, though the FactCheck-GPT's reliance on parametric knowledge limits its reliability for new or updated information. Appendix 10.11 also provides a case study of how different methods handle challenging content units in LM responses to `Hard` prompts.

7.3 Refusal Rate Impacts LM Factuality

The current factual precision metric (detailed in 10.8) does not evaluate LM factuality when models refrain from answering. In this section, we investigate LMs' refusal rate and its significance. Previous work (Min et al., 2023) relied on simple heuristics or keywords to identify refusal responses, but our initial investigation finds these methods unreliable. To address this, we prompt GPT-4 Turbo to detect refusals and categorize them into different types, including clarification requests, lack of knowledge, safety concerns, and misinformation risks. Figure 5 shows the refusal rates for each model across different FACTBENCH tiers (see Appendix 10.12 for the task prompt and refusal category distributions across LMs). Notably, Gemini1.5-Pro exhibits a significantly higher refusal rate than other models, approaching 10% on the `Hard` portion of FACTBENCH. As shown, it refuses to answer prompts in the `Hard` tier more frequently which contains the most challenging hallucination prompts. Moreover, while Gemini1.5-Pro's refusal strategies can help prevent hallucinations (see example 4), the high rate of refusals impacts its overall factuality precision. Additionally, our manual inspection reveals that in

25% refusal cases, Gemini1.5-Pro's reason for not answering is not valid. For example, the model interpreted "give me studies on the recommended interval between COVID vaccines" as a request for medical advice and refused to answer. Our analysis of unnecessary refusals reveals that Gemini1.5-Pro overestimates risks in two categories: 29% involve ethical and legal queries, while 49% (mostly `Hard` prompts) stem from perceived misinformation risks, indicating overly conservative content filtering. These findings highlight the need for more nuanced factuality evaluation metrics to account for refusals as an important future direction for future research.

8 Conclusion

We present VERIFY, a factuality evaluation pipeline that annotates LM responses in real-world settings by decomposing them into content units and labeling them as `supported`, `unsupported`, or `undecidable` based on Web evidence. Our method shows a stronger correlation with human evaluations compared to existing approaches. Using VERIFY, we curated FACTBENCH, a 1,000-prompt benchmark across 150 topics, organized into `Hard`, `Moderate`, and `Easy` tiers. As the first updatable real-world factuality benchmark, FACTBENCH enables tracking evolving challenges in LM factuality. Our evaluation of frontier LMs reveals superior performance of proprietary models, with factual precision declining from `Easy` to `Hard` tiers.

9 Limitations

Similar to previous factuality evaluation approaches (Wang et al., 2024a; Wei et al., 2024b; Min et al., 2023), VERIFY employs a single language model for text decomposition and unit annotation. Extending the framework to leverage multiple LMs could enhance evaluation diversity and mitigate individual model biases. However, this expansion raises several challenges: (i) determining whether to maintain a single model for initial text decomposition while incorporating multiple models for verification, (ii) designing effective inter-model collaboration mechanisms, and (iii) balancing enhanced robustness against computational costs. Future work should systematically address these methodological and computational trade-offs to develop more comprehensive multi-model factuality evaluation systems. Another limitation of our work is the absence of recall measurements - a challenge particularly salient for open-ended queries. For example, defining an exhaustive set of relevant factual statements in movie recommendation scenarios is inherently difficult, as models may produce accurate but incomplete information. While our method strongly correlates with human judgments on evaluated statements, addressing this recall challenge remains crucial for high-stakes applications. Future work should also explore evaluation pipelines that consider both individual factual support and logical connections between units, verifying not only factual precision but also response-wide coherence.

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10 Appendix

10.1 Data Cleaning

We begin by collecting prompts from the first turn of conversations in the LMSYS-Chat-1M dataset, which is a large-scale, in-the-wild LM conversations dataset. Since the existing language labels are unreliable, we employ the Llama3-70B-Instruct model (AI@Meta, 2024) to identify the language of each conversation using the prompt in Appendix 10.15.1. This gives us 516,771 distinct English prompts with at least 32 characters. Next, we remove anonymized (30.9%) and duplicated (12.1%) prompts. Meanwhile, we observed that some users queried LMs with thousands of identical prompts. To mitigate this issue's impact on subsequent clusters, we filter out prompts with a Jaccard similarity score greater than 0.9. Our cleaned data contains 294,333 distinct prompts.

10.2 Manual Check on Prompt Verifiability

In order to ensure the verifiability specified in Section 3, three authors have conducted multiple rounds of human inspection and validation to exclude all non-verifiable prompts like problem-solving (e.g., "A suit manufacturer has 14 suits for men and 4 suits for women. How many suits are available overall?") and faithfulness-related (e.g., "Translate the given text") tasks. More unverifiable examples are available in our prompt at Appendix 10.15.2.

10.3 BERTopic Parameter Tuning

According to the BERTopic guideline, only the parameters of the clustering stage significantly impact topic modeling quality. Therefore, we used default settings for other stages including sentence-to-vector mapping and dimensionality reduction, and used a grid search to explore combinations of key HDBSCAN parameters: `min_cluster_size` (set to 100, the minimum number of prompts per cluster) and `min_samples` (set to 25, the density threshold for outlier detection). The grid search spanned values of 10, 25, 50, 100, and 200 for both parameters. We evaluated clustering quality through manual inspection, focusing on topic granularity across the top and bottom 50 clusters while avoiding overly specific topics (e.g., "Taylor Swift's birthday") or overly general ones (e.g., "question-answering").

Our manual inspection showed that clustering results remained stable across different parameter combinations, with only marginal improvements at the chosen values of `min_cluster_size=100` and `min_samples=25`. This robustness aligns with HDBSCAN's reputation for requiring minimal tuning, making it well-suited for real-world applications with limited prior knowledge of the underlying data structure.

10.4 Usefulness Scoring Details

Through multiple rounds of discussion and empirical testing, we define the criteria to capture the most critical aspects of prompt usefulness as follows:

- **Clarity**: This criterion assesses whether the prompt is easily understandable and is not ambiguous.
- **Generalizability**: We developed this criterion to prevent over-specialization. The assessment focuses on the prompt's potential to be meaningful across different contexts or users.
- **Relevance**: This criterion assesses whether the information requested is important and potentially interesting to a broader audience.
- **Feasibility**: This criterion evaluates whether the requested information is reasonably provided within the LM's capabilities.

Our scoring methodology involved two frontier LMs (GPT-4 Turbo and Llama3-70B-Instruct) independently scoring each criterion on a scale from 0 (lowest) to 5 (highest). The aggregate score calculation leverages a formula that balances multiple models' perspectives:

$$S(P) = \frac{1}{|C|} \sum_{c \in C} \sum_{m \in M} S_m(c)$$

where C denotes the set of criteria {clarity, generalizability, relevance, feasibility}, M denotes the set of models {*GPT-4 Turbo*, *Llama3-70B-Instruct*}, and $S_m(c)$ denotes the score that model m assigns to criterion c. This approach reduces individual model bias and ensures a comprehensive evaluation of prompt usefulness, allowing us to create a more robust and reliable dataset for further research and analysis. The scoring prompt is provided in Appendix 10.15.3.

10.5 α Tuning for Hallucination Score

The weighting factor α in equation 1 balances the importance of undecidable and unsupported units.

To determine the appropriate α value, we analyzed 100 responses (25 per model). Two annotators evaluated 570 undecidable units, achieving strong inter-annotator agreement (85.5%). Across all models, 57% of undecidable units were found to be factual and 43% not factual, with individual models showing similar patterns as shown in Table 2. Based on this finding, we set $\alpha = 0.5$.

10.6 FACTBENCH's Updating Process

FACTBENCH identifies prompts within the LMSYS-chat-1M dataset (Zheng et al., 2024) that challenge LMs in factual generation. We plan to annually incorporate new prompts from the LMSYS-chat-1M dataset, which the authors intend to release quarterly. Our future work also includes expanding our prompt collection by identifying hallucination prompts from the WildChat dataset (Zhao et al., 2024), another rich source of user-model interactions with regular updating of the conversations.

For new interaction data from subsequent years, we apply the Collect pipeline (Section 3) to identify representative prompt clusters, followed by evaluating these prompts for Verifiability and Usefulness using established parameters and methods. This process generates a new set of candidate prompts. Next, we need to combine the resulting prompts with the existing FACTBENCH prompts, where we face two challenges:

- Old prompts may overlap with the new ones. To address this issue, we remove existing prompts if they fall into clusters covered by new candidates.
- As proprietary models are continuously updated, we regenerate responses for existing prompts using the latest model versions to ensure they remain challenging.

After addressing these two challenges, we obtain a combined prompt set with current model responses. We then apply the VERIFY pipeline to compute hallucination scores, using these rankings, as well as out tiered approach, to curate the next version of FACTBENCH.

10.7 Baselines Description

We use `gpt-3.5-turbo-0613` (Brown et al., 2020) as a backbone LM when running all baselines.

- **FactScore** (Min et al., 2023): FactScore evaluates the factual precision of LMs by breaking text into atomic facts and assessing the percentage of facts supported by Wikipedia articles. The original FactScore method is provided with Wikipedia pages with relevant information. However, the extracted units from in-the-wild requests are not associated with a Wikipedia page and might not even be found in Wikipedia articles. To make a fair comparison, we use the Wikipedia API (Goldsmith, 2014) to map these atomic units to the 5 closest Wikipedia topics in the Wiki database for retrieval.
- **Search-Augmented Factuality Evaluator (SAFE)** (Wei et al., 2024b): SAFE evaluates long-form factuality by decomposing text into atomic facts, adopting the same FactScore fact extraction component, and checking each fact's relevancy to the original query. For relevant facts, SAFE queries the Google search engine for evidence retrieval and labels each fact as either supported or refuted accordingly.
- **Factcheck-GPT** (Wang et al., 2024a): Factcheck-GPT is a hallucination detection and mitigation framework. In the annotation phase, it assesses the factuality of LM-generated content using a multi-step annotation pipeline that includes the decomposition of claims, decontextualization, evidence retrieval through Google Search, evidence snippets generation, final factuality decision, and revision of non-factual elements. For this study, the final revision step is excluded from the baseline methodology.

10.8 Factual Precision Metric

We adopt the factual precision utilized by FactScore (Min et al., 2023) to compare the performance of different models on FACTBENCH. Given the set of prompts P and knowledge source K, we first obtain model M responses $\{R_M = M(p) \text{ for } p \in P\}$. All baselines decompose each response into atomic units (facts). Therefore, we denote U to be the set of units in R_M. We calculate the **factual precision** of R_M as:

$$f(R_M) = \frac{1}{|U|} \sum_{u \in U} \mathbb{I}[u \text{ is supported by } K] \quad (2)$$

The overall factuality precision of each model on P prompts is calculated as:

$$F(M) = \mathbb{E}_{p \in P}[f(M_p | M_p \text{ responds}] \quad (3)$$

Model	Factual (Avg. Percentage)	Not Factual (Avg. Percentage)
GPT4-o	68.4	31.6
Gemini1.5-Pro	56.6	43.4
Llama3.1-405B-Instruct	51.0	49.0
Llama3.1-70B-Instruct	52.0	48.0
Average	**57.0**	**43.0**

Table 2: Distribution of manually-annotated factual and non-factual `undecidable` units across LMs.

We do not evaluate factual recall, such as cases where models abstain from answering or generate responses with minimal factual content. Wei et al. (2024b) suggests that there is a fixed number of content units users care about and that this number can be tuned. However, we did not find this metric compelling, as different models show different verbosity levels, making it difficult to establish a unified threshold. Moreover, the number of units is not a reliable quality indicator, as the content may still lack relevance or usefulness.

10.9 Accuracy of Factuality Evaluation Methods across LMs

Figure 7 (left) evaluates the ability of different methods to verify *Independent* units per LM. However, factuality evaluation methods often generate both dependent and independent units, requiring mechanisms to handle ambiguous units that are rather difficult to verify using existing online knowledge. Figure 7 (right) evaluates the ability of different methods to verify all kinds of units (including both *Independent* and *Dependent* units). By comparing the right figure with the left one, we can evaluate the robustness of factuality evaluation methods when handling ambiguous and contextually dependent units. As shown in the right figure, the gap between VERIFY and the best-performing baseline, Factcheck-GPT, narrows to an average accuracy difference of 0.6%. This suggests VERIFY 's improved accuracy relative to other baselines when dealing with dependent units.

10.10 Leaderboard

Table 3 presents a comprehensive list of benchmarked LMs, including the four models discussed in the main paper and three additional models: Claude-3.5-Sonnet, CommandR+, and Mistral-Large-2.

10.11 Qualitative analysis of VERIFY in annotating challenging units

Here is a qualitative analysis that demonstrate VERIFY's handling of difficult judgments, comparing its evaluation results with FactCheck-GPT on responses generated by Llama3.1-405B-Instruct. We focus on comparing with FactCheck-GPT since other baselines showed significantly weaker performance.

Prompt:
What are the best solutions for quickly and easily testing prompts with multiple different LLM models/services?
Response:
1. **LLaMA**: LLaMA (Large Language Model Application) is an open-source framework that allows you to test prompts with multiple LLM models, including those from Hugging Face, Google, and Microsoft. LLaMA provides a simple API for testing prompts and supports multiple input formats.
2. **Prompt Engineering**: Prompt Engineering is a platform that allows you to design, test, and optimize prompts for multiple LLM models. It provides a visual interface for crafting prompts and supports testing with models from Hugging Face, Google, and Microsoft.

- **Unit 1: LLaMA supports models from Hugging Face, Google, and Microsoft.**
  - **FactCheck-GPT:** Supported
  - **VERIFY:** Unsupported
  - **Analysis:** VERIFY correctly identified this technical inaccuracy, as LLaMA is a standalone model family that does not integrate with these platforms.
- **Unit 2: LLaMA supports multiple input formats.**
  - **FactCheck-GPT:** Supported
  - **VERIFY:** Undecidable

	Model	Factual Precision	Hallucination Score
Hard	GPT4-o	**75.65**	0.64
	Gemini1.5-Pro	73.78	0.68
	Llama3.1-70B-Instruct	70.07	0.89
	Llama3.1-405B-Instruct	68.59	0.93
	Claude-3.5-Sonnet	74.95	0.65
	CommandR+	73.15	0.71
	Mistral-Large-2	75.19	0.67
Moderate	GPT4-o	**80.72**	0.50
	Gemini1.5-Pro	78.02	0.57
	Llama3.1-70B-Instruct	75.76	0.71
	Llama3.1-405B-Instruct	75.05	0.70
	Claude-3.5-Sonnet	79.92	0.54
	CommandR+	80.71	0.52
	Mistral-Large-2	79.97	0.52
Easy	GPT4-o	91.63	0.26
	Gemini1.5-Pro	89.86	0.31
	Llama3.1-70B-Instruct	89.30	0.33
	Llama3.1-405B-Instruct	86.57	0.40
	Claude-3.5-Sonnet	89.61	0.30
	CommandR+	91.65	0.25
	Mistral-Large-2	**92.00**	0.25

Table 3: Leaderboard of LMs benchmarked on FACTBENCH. The best model performance in each prompt category is in **bold**. Updated and detailed benchmarking results can be found at our project page: https://huggingface.co/spaces/launch/factbench

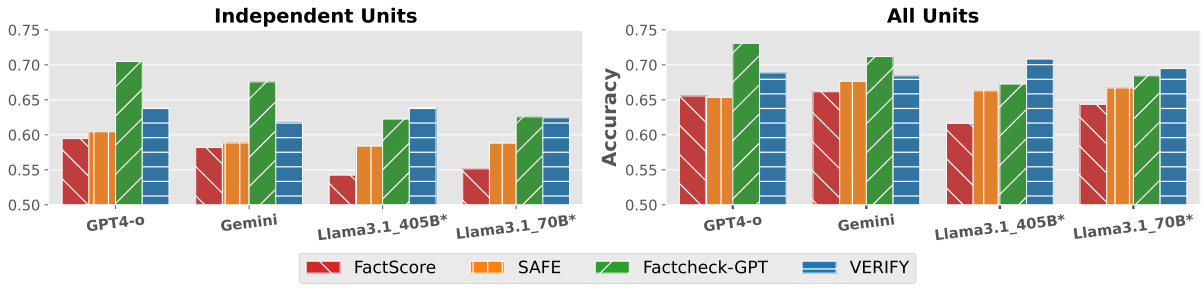


Figure 7: Accuracy of factuality evaluation labeling compared to human annotations across LMs. **Independent Units** are independent units identified by human annotators, and **All Units** include both dependent and independent units. VERIFY is robust to dependent units, as demonstrated by the noticeable accuracy improvement in the right figure.

  - **Analysis:** VERIFY appropriately flagged this as undecidable since input format support varies across LLaMA versions and implementations.

- **Unit 3: Prompt Engineering is a platform that allows you to design, test, and optimize prompts for multiple LLM models.**
  - **FactCheck-GPT:** Supported
  - **VERIFY:** Contradicted
  - **Analysis:** VERIFY correctly identified that prompt engineering is a methodology, not a platform, showing its ability to distinguish conceptual differences.

- **Unit 4: Prompt Engineering supports testing with models from Hugging Face, Google, and Microsoft.**
  - **FactCheck-GPT:** Supported
  - **VERIFY:** Undecidable
  - **Analysis:** VERIFY correctly labeled this as undecidable since prompt engineering, as a methodology, can be applied to any model without having explicit *support*.

10.12 Refusal Prompt and Refusal Type Distributions

The refusal categories explain various reasons for declining to answer queries. "No Refusal" indicates a complete response, while categories like "Safety Concerns" and "Misinformation Risks" reflect avoidance of harmful or misleading information. Refusals may also stem from requests for "Sensitive or Private Information," where personal data is involved, or a "Clarification Request," where the model seeks further details. Other reasons include "Ethical and Legal Advice," "Hate Speech or Discrimination," and "Lack of Knowledge/Capability," which acknowledge the model's limitations. The "Other" category covers refusals that don't fit these reasons.

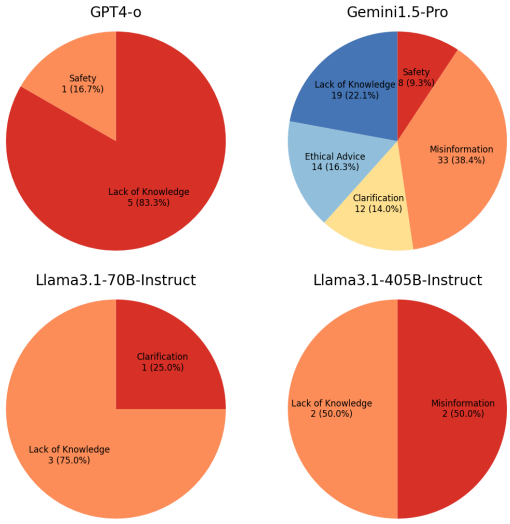


Figure 8: Distribution of LM's answer refusal categories on FACTBENCH

Figure 8 shows the distribution of refusal categories across LMs, with Gemini1.5-Pro having the highest number of refusals across a broad range of categories. We also show an example in Table 4 where Gemini1.5-Pro refrained from answering by requesting clarifications while GPT4-o produced a fully hallucinated response.

Model	Model Response
GPT4-o	Sure! Fuzhou, the capital of China's Fujian province, is known for its rich culinary culture. Here are some highly recommended restaurants where you can try a variety of delicious dishes: 1. **Ding Feng Zhuang**: - *Cuisine*: Traditional Fujian ... 2. **Huangxiang Shousibao**: - *Cuisine*: Seafood ... 3. **Yu Xiangyan**: - *Cuisine*: Fujian and Cantonese... 4. **Juchunyuan** : - *Cuisine*: Fujian... 5. **Shawanrong**: - *Cuisine*: Modern Fusion...
Gemini1.5-Pro	I can definitely recommend some restaurants in Fuzhou City! To give you the best suggestions, please tell me: - What kind of cuisine are you interested in? (e.g., Fuzhou specialties, Cantonese, Sichuan, Western, etc.) - What's your budget like? (e.g., fine dining, mid-range, budget-friendly) - What's your dining style? (e.g., street food, casual, formal, romantic) - What area of Fuzhou are you interested in? (Specific neighborhoods or landmarks will help!) Once I have this information, I can provide you with a more tailored and helpful list of restaurant recommendations.
Llama3.1-70B-Instruct	Fuzhou, the capital of Fujian Province in China, is known for its rich culinary heritage and delicious local dishes. Here are some recommended restaurants in Fuzhou City: 1. **Buddha Jumps Over the Wall**: This restaurant is famous for its signature dish, "Buddha Jumps Over the Wall," a traditional Fujianese soup ... 2. **Fuzhou Fish Ball Restaurant**: This casual eatery is a local favorite for its delicious fish balls, which are made from fresh fish and served in a light broth. 3. **Gongdelin Vegetarian Restaurant**: This vegetarian restaurant offers a wide range of plant-based dishes, including traditional Fujianese options like stir-fried vegetables and tofu...
Llama3.1-405B-Instruct	Fuzhou, the capital of Fujian Province in China, is renowned for its rich culinary heritage... Here are some recommended restaurants in Fuzhou that showcase the best of Fujian cuisine and more: 1. **Dongbai Lou** - Known for its traditional Fujian dishes, including the famous Dongbai chicken and a variety of seafood options. ... 2. **Wu Ming Lou** - Offers a mix of Fujian and Cantonese dishes ... 3. **Fuzhou Wanshou Road Food Street** - While not a single restaurant, this food street is a must-visit for any food enthusiast... 4. **Min Jiang Lou** - Specializes in Fujian cuisine with a focus on river and sea delicacies... 5. **Huangcheng Hotel Restaurant** - Offers a range of Fujian dishes in a comfortable setting...

Table 4: Varying behavior of LMs when confronted with knowledge limitations. GPT4-o, Llama3.1-70B-Instruct, and Llama3.1-405B-Instruct produce hallucinated or inaccurate content (marked in red), while Gemini1.5-Pro either refrains from generating a response or asks for clarifications to better understand the query: *"Recommend some restaurants in Fuzhou City"*.

Below is the prompt used for refusal detection and categorization:

```
Evaluate whether the following RESPONSE from a large
      language model contains a refusal to answer and
      categorize the type of refusal.

0. No Refusal: The response does not contain a refusal.
1. Safety Concerns: The response avoids answering due to
      harmful, dangerous, or illegal content in the query.
2. Misinformation Risks: The model refuses because
      answering might spread false or misleading
      information, especially in critical areas (e.g.,
      medical, legal, or financial advice).
3. Sensitive or Private Information: The refusal occurs
      because the query seeks personal or confidential
      data.
4. Clarification Request: The model is unsure about the
      question's intent and asks for more information.
5. Ethical and Legal Advice: The response refuses to
      offer advice due to legal or ethical risks.
6. Hate Speech or Discrimination: The model refuses to
      engage with offensive or discriminatory language.
7. Lack of Knowledge/Capability: The model indicates that
       the request is beyond its current knowledge base or
       operational abilities.
8. Other: The refusal does not fit into any of the above
      categories.

Use the following JSON object format for your evaluation:
{
    "Annotatation": "yes/no",
    "Categorization": 0-8,
    "Explaination": "brief explanation for the refusal
          type"
}

RESPONSE: [MODEL_RESPONSE]
```

10.13 Benchmark Topic Distribution

Figure 9 illustrates the percentage of verifiable prompts across the top 50 topic clusters. Table 5 demonstrates prompt statistics in each tier.

	Models	# Prompts	# Selected Prompts	Total Prompts	Total Selected Prompts
Hard	gpt-4	3431	500	15499	2205
	claude-2	1074	181		
	gpt-3.5-turbo	3607	524		
	claude-1	7387	1000		
Moderate	claude-instant-1	2422	171	30613	1435
	vicuna-33b	10548	434		
	llama-2-13b-chat	12160	628		
	wizardlm-13b	5483	202		
Easy	mpt-30b-chat	3150	11	195641	542
	vicuna-13b	183117	500		
	palm-2	2463	8		
	guanaco-33b	5282	20		
	llama-2-7b-chat	1629	3		

Table 5: Prompt statistics of LMs in each Tier (**Hard**, **Moderate**, **Easy**).

10.14 Case Study on LLaMA3.1-405B-Instruct and LLaMA3.1-70B-Instruct performance gap

As mentioned in the main text, the performance of Llama3.1-405B-Instruct on VERIFY is close to or even lower than that of Llama3.1-70B-Instruct. We conduct an in-depth investigation into the possible reasons. Based on results in Figure 3, we find that Llama3.1-405B-Instruct has the highest proportion of `undecidable` labels among all the models, directly resulting in its lower factual precision. After manually reviewing 30 cases with the largest factual precision gap between the 405B and 70B models, we observed a common phenomenon: the 405B model tends to use more subjective modifiers such as "solid", "exclusive", and "well-known", while the 70B model usually prefer plain descriptions. For example, consider the following case when both 405B and 70B models respond with the same Debian guidebook:

Llama3.1-405B-Instruct
Atomic Unit: "Debian GNU/Linux: Guide to Installation and Usage" by John Goerzen and Ossama Othman is a solid foundation for understanding Debian's basics.
Reasoning: One KNOWLEDGE point states that the book is a "nice book and CD combo" that can get someone started with Debian, but notes that the book lacks in-depth information, While the book is mentioned as a "nice" resource for getting started with Debian, it is also criticized for lacking in-depth information. Therefore, the evidence is insufficient to confirm or deny that the book is a solid foundation.
Label: `undecidable`

Llama3.1-70B-Instruct:
Atomic Unit: "Debian GNU/Linux: Guide to Installation and Usage" by John Goerzen and Ossama Othman is a guidebook of Debian basics.
Reasoning: The KNOWLEDGE points strongly imply that the book is a guide that covers basics of Debian.
Label: `suppported`

Based on the case above, we find that due to introducing a detailed CoT reasoning mechanism and the `undecidable` label, VERIFY is very rigorous in verifying atomic units, especially when subjective adjectives are involved. We use a pretrained subjectivity classification model `GroNLP/mdebertav3-subjectivity-multilingual` (Ruggeri et al., 2023) to classify the output units of both

models. The classification results show that 9.4% of the 405B models response units are classified as subjective, while only 7.5% of the 70B models response units are subjective. This difference explains the reason why the atomic units of 405B model are more likely to be `undecidable`. Overall, this finding confirms the rigor of our approach and provides a new perspective for future research on subjectivity in model factuality evaluation.

10.15 Prompts

In this section, we show the prompts we used throughout the experiments.

10.15.1 Language Detection

```
Determine if the following input sentence is English or
      not. Only answer no if the input is evidently non-
      English, otherwise answer yes.

Input: Please translate "How are you today" to Spanish.
Your Answer: yes

Input: OK
Your Answer: yes

Input: Ecco dieci frasi in italiano che potresti
Your Answer: no

Input: I
Your Answer: yes

Input: Answer: D
Your Answer: yes

Input: negative
Your Answer: yes

Input: En fran\c{c}ais, on dirait: "La douleur est in\'{e
      }vitable, la souffrance est un choix".
Your Answer: no

Input: {user_prompt}
Your Answer:
```

10.15.2 Factual Prompt Labeling

```
Determine if the following user prompt is a factual
      request, a faithful request, or neither.
Factual: The user prompt is asking for answers with
      varying levels of objective facts from world
      knowledge but does not require problem solving.
Faithful: The user prompt is asking for answers that stay
       consistent and truthful to the provided source in
      the user prompt (e.g., data-to-text, translation).
Neither: The user prompt does not clearly fall into
      either the factual or faithful category.
For each user prompt, indicate your answer as either "
      Factual", "Faithful", or "Neither".

User prompt: Who won the last World Cup of football?
Your Answer: Factual

User prompt: what functional groups does C/C=C/c2ccc(
      COc1cccc(CCO)c1)cc2 contain?
Your Answer: Neither

User prompt: Please translate "How are you today" to
      Spanish.
Your Answer: Faithful

User prompt: From now on you will roleplay as my wife.
Your Answer: Neither

User prompt: What's the difference between GitHub and Git
      .
Your Answer: Factual

User prompt: A suit manufacturer has 14797 suits for men
      and 4969 suits for women. How many suits are
      available overall?
Your Answer: Neither

User prompt: Convert the following temperature from
      Celsius to Fahrenheit: 25°C.
Your Answer: Faithful

User prompt: Generate a code to find all prime numbers in
       from 0 to 100k
Your Answer: Neither

User prompt: Can you write me a blog post about George
      Washington?
Your Answer: Factual

User prompt: write a story about a cat that meowed all
      the time
Your Answer: Neither

User prompt: {user_prompt}
Your Answer:
```

10.15.3 Prompt Usefulness Scoring

```
Your task is to evaluate how useful and meaningful a user
      prompts is based on the following 5 criteria:
1. Clarity (0-5): Is the prompt easily understandable
     without leaving any ambiguity?
2. Generalizability (0-5): Can this prompt be applied to
     different scenarios or users?
3. Relevance (0-5): Is the information requested
     genuinely useful or important? Does it have
     potential interest/value to a broader audience?
4. Actionability (0-5): Is the information requested
     likely to inform decisions or trigger actions? Does
     it have practical implications?
5. Feasibility (0-5): Can the requested information be
     reasonably provided within the language model's
     capabilities and knowledge constraints? Is it asking
      for information that exists and is accessible?

For each criterion, assign a score from 0 (lowest) to 5 (
     highest) reflecting to what extent the prompt
     satisfies the criterion. \
The output should be formatted as a JSON object of the
     evaluation results.

Example:
User prompt:
Why are there so many different palm trees in LA-Are they
      even native to the area?

Evaluation Results:
{"Clarity": 4, "Generalizability": 2, "Relevance": 3, "
     Actionability": 2, "Feasibility": 5}

Your Task:
User prompt:
[USER_PROMPT]

Evaluation Results:
```

10.15.4 Unit Extraction Prompt

```
Instructions:
- Exhaustively break down the following text into
     independent content units. Each content unit can
     take one of the following forms:
  a. Fact: An objective piece of information that can be
       proven or verified.
  b. Claim: A statement or assertion that expresses a
       position or viewpoint on a particular topic.
  c. Instruction: A directive or guidance on how to
       perform a specific task.
  d. Data Format: Any content presented in a specific
       format, including code, mathematical notations,
       equations, variables, technical symbols, tables,
       or structured data formats.
  e. Meta Statement: Disclaimers, acknowledgments, or any
        other statements about the nature of the response
        or the responder.
  f. Question: A query or inquiry about a particular
       topic.
  g. Other: Any other relevant content that doesn't fit
       into the above categories.
- Label each content unit with its corresponding unit
     type using the format: [content unit]: [content unit
      type]
- Refer to the following examples to understand the task
     and output formats.

Example 1:
TEXT: Zhejiang Huafang Pharmaceutical Co., Ltd. is a
     leading chemical company based in China that
     specializes in the research, manufacturing, and
     sales of various pharmaceutical products, including
     excipients and intermediates. The company was
     founded in 2018 and is located in Hangzhou, a city
     with a rich history in eastern China. Zhejiang
     Huafang Pharmaceutical Co., Ltd. is committed to
     providing high-quality products to its customers in
     the healthcare industry. The company's manufacturing
      facilities are equipped with state-of-the-art
     technology and infrastructure that ensure the
     production of high-quality products. Overall,
     Zhejiang Huafang Pharmaceutical Co., Ltd. is a
     reputable pharmaceutical company with a long history
      of success in the healthcare industry. The company'
     s commitment to quality, innovation, and customer
     service has made it a leader in the field of
     pharmaceutical research and development.

UNITS:
- Zhejiang Huafang Pharmaceutical Co., Ltd. is a leading
     chemical company: Fact
- Zhejiang Huafang Pharmaceutical Co., Ltd. is based in
     China: Fact
- Zhejiang Huafang Pharmaceutical Co., Ltd. specializes
     in the research of various pharmaceutical products:
     Fact
- Zhejiang Huafang Pharmaceutical Co., Ltd. specializes
     in the manufacturing of various pharmaceutical
     products: Fact
- Zhejiang Huafang Pharmaceutical Co., Ltd. specializes
     in the sales of various pharmaceutical products:
     Fact
- excipients are the pharmaceutical products of the
     Zhejiang Huafang Pharmaceutical Co., Ltd.: Fact
- intermediates are the pharmaceutical products of the
     Zhejiang Huafang Pharmaceutical Co., Ltd.: Fact
- The company was founded in 2018: Fact
- The company is located in Hangzhou: Fact
- Hangzhou is a city: Fact
- Hangzhou has a rich history in eastern China: Fact
- Zhejiang Huafang Pharmaceutical Co., Ltd. is committed
     to providing high-quality products to its customers
     in the healthcare industry: Claim
- The company's manufacturing facilities are equipped
     with state-of-the-art technology: Fact
- The company's manufacturing facilities are equipped
     with state-of-the-art infrastructure: Fact
- The company's manufacturing facilities are equipped
     with state-of-the-art technology and infrastructure
     that ensure the production of high-quality products:
      Claim
- Zhejiang Huafang Pharmaceutical Co., Ltd. is a
     reputable pharmaceutical company: Claim
- Zhejiang Huafang Pharmaceutical Co., Ltd. has a long
     history of success in the healthcare industry: Claim
- The company is committed to quality: Claim
- The company is committed to innovation: Claim
- The company is committed to customer service: Claim
- The company's commitment to quality, innovation, and
     customer service has made it a leader in the field
     of pharmaceutical research: Claim
- The company's commitment to quality, innovation, and
     customer service has made it a leader in the field
     of pharmaceutical development: Claim

Example 2:
TEXT: I'm here to help you make an informed decision.
     Both the RTX 3060 Ti and RTX 3060 are powerful GPUs,
      and the difference between them lies in their
     performance. The RTX 3060 Ti has more CUDA cores
     (4864 vs 3584) but a lower boost clock speed (1665
     MHz vs 1777 MHz) compared to the RTX 3060. In terms
     of memory bandwidth, the RTX 3060 Ti has a slight
     edge over the RTX 3060 with a bandwidth of 448 GB/s
     compared to 360 GB/s. However, the difference is
     relatively small. It's important to consider other
     factors such as the power consumption, cooling
     system, and compatibility with your system when
     making a decision."

UNITS:
- I'm here to help you make an informed decision: Meta
     Statement
- The RTX 3060 Ti is a powerful GPU: Claim
- The RTX 3060 is a powerful GPU: Claim
- The difference between them lies in their performance:
     Claim
- The RTX 3060 Ti has more CUDA cores compared to the RTX
      3060: Fact
- The RTX 3060 Ti has 4864 CUDA cores: Fact
- The RTX 3060 has 3584 CUDA cores: Fact
- The RTX 3060 Ti has a lower boost clock speed compared
     to the RTX 3060: Fact
- The RTX 3060 Ti has a boost clock speed of 1665 MHz:
     Fact
- The RTX 3060 has a boost clock speed of 1777 MHz: Fact
- The RTX 3060 Ti has a slight edge over the RTX 3060 in
     terms of memory bandwidth: Fact
- The RTX 3060 Ti has a memory bandwidth of 448 GB/s:
     Fact
- The RTX 3060 has a memory bandwidth of 360 GB/s: Fact
- The difference is relatively small: Claim
- It's important to consider other factors such as power
     consumption when making a decision: Instruction
- It's important to consider other factors such as
     cooling system when making a decision: Instruction
- It's important to consider other factors such as
     compatibility with your system when making a
     decision: Instruction

Your Task:
TEXT: {_RESPONSE_PLACEHOLDER}
UNITS:
```

10.15.5 Decontextualization Prompt

```
You task is to decontextualize a UNIT to make it
     standalone. \
Each UNIT is an independent content unit extracted from
     the broader context of a RESPONSE.
```

```
Vague References:
- Pronouns (e.g., "he", "she", "they", "it")
- Demonstrative pronouns (e.g., "this", "that", "these",
      "those")
- Unknown entities (e.g., "the event", "the research", "
      the invention")
- Incomplete names (e.g., "Jeff..." or "Bezos..." when
      referring to Jeff Bezos)

Instructions:
Follow the steps below for unit decontextualization:
1. If the UNIT contains vague references, minimally
      revise them with respect to the specific subjects
      they refer to in the RESPONSE.
2. The decontextualized UNIT should be minimally revised
      by ONLY resolving vague references. No additional
      information must be added.
3. UNIT extraction might decompose a conjunctive
      statement into multiple units (e.g. Democracy treats
       citizens as equals regardless of their race or
      religion -> (1) Democracy treats citizens as equals
      regardless of their race, (2) Democracy treats
      citizens as equals regardless of their religion).
      Avoid adding what is potentially part of another
      UNIT.
4. Provide a reasoning of the revisions you made to the
      UNIT, justifying each decision.
5. After showing your reasoning, provide the revised unit
       and wrap it in a markdown code block.

Example 1:
UNIT:
Acorns is a financial technology company

RESPONSE:
Acorns is a financial technology company founded in 2012
      by Walter Cruttenden, \
Jeff Cruttenden, and Mark Dru that provides micro-
      investing services. The \
company is headquartered in Irvine, California.

REVISED UNIT:
This UNIT does not contain any vague references. Thus,
      the unit does not require any further
      decontextualization.
```
Acorns is a financial technology company
```

Example 2:
UNIT:
The victim had previously suffered a broken wrist.

RESPONSE:
The clip shows the victim, with his arm in a cast, being
      dragged to the floor \
by his neck as his attacker says "I'll drown you" on a
      school playing field, while forcing water from a
      bottle into the victim's mouth, \
simulating waterboarding. The video was filmed in a lunch
       break. The clip shows the victim walking away,
      without reacting, as the attacker \
and others can be heard continuing to verbally abuse him.
       The victim, a Syrian refugee, had previously
      suffered a broken wrist; this had also been \
investigated by the police, who had interviewed three
      youths but took no further action.

REVISED UNIT:
The UNIT contains a vague reference, "the victim." This
      is a reference to an unknown entity, \
since it is unclear who the victim is. From the RESPONSE,
       we can see that the victim is a Syrian refugee. \
Thus, the vague reference "the victim" should be replaced
       with "the Syrian refugee victim."
```
The Syrian refugee victim had previously suffered a
      broken wrist.
```

Example 3:
UNIT:
The difference is relatively small.

RESPONSE:
Both the RTX 3060 Ti and RTX 3060 are powerful GPUs, and
      the difference between them lies in their
      performance. \
The RTX 3060 Ti has more CUDA cores (4864 vs 3584) but a
      lower boost clock speed (1665 MHz vs 1777 MHz)
      compared to the RTX 3060. \
In terms of memory bandwidth, the RTX 3060 Ti has a
      slight edge over the RTX 3060 with a bandwidth of
      448 GB/s compared to 360 GB/s. \
However, the difference is relatively small and may not
      be noticeable in real-world applications.

REVISED UNIT:
The UNIT contains a vague reference, "The difference."
      From the RESPONSE, we can see that the difference is
       in memory bandwidth between the RTX 3060 Ti and RTX
       3060. \
Thus, the vague reference "The difference" should be
      replaced with "The difference in memory bandwidth
      between the RTX 3060 Ti and RTX 3060." \
The sentence from which the UNIT is extracted includes
      coordinating conjunctions that potentially decompose
       the statement into multiple units. Thus, adding
      more context to the UNIT is not necessary.
```
The difference in memory bandwidth between the RTX 3060
      Ti and RTX 3060 is relatively small.
```

YOUR TASK:
UNIT:
{UNIT}

RESPONSE:
{RESPONSE}

REVISED UNIT:
```

10.15.6 Query Generator Prompt

```
Instructions:
You are engaged in a multi-round process to refine Google
       Search queries about a given STATEMENT. \
Each round builds upon KNOWLEDGE (a list of previous
      queries and results, starting empty in round 1). \
Your goal is to improve query quality and relevance over
      successive rounds.

QUERY CONSTRUCTION CRITERIA: a well-crafted query should:
  - Retrieve information to verify the STATEMENT's
       factual accuracy.
  - Seek new information not present in the current
       KNOWLEDGE.
  - Balance specificity for targeted results with breadth
        to avoid missing critical information.
  - In rounds 2+, leverage insights from earlier queries
       and outcomes.

Process:
1. Construct a Useful Google Search Query:
  - Craft a query based on the QUERY CONSTRUCTION
       CRITERIA.
  - Prioritize natural language queries that a typical
       user might enter.
  - Use special operators (quotation marks, "site:",
       Boolean operators, intitle:, etc.) selectively and
        only when they significantly enhance the query's
       effectiveness.

2. Provide Query Rationale (2-3 sentences):
  Explain how this query builds upon previous efforts and
       /or why it's likely to uncover new, relevant
       information about the STATEMENT's accuracy.

3. Format Final Query:
  Present your query in a markdown code block.

KNOWLEDGE:
{_KNOWLEDGE_PLACEHOLDER}

STATEMENT:
{_STATEMENT_PLACEHOLDER}
```

10.15.7 Final Accuracy Decision Prompt

```
Instructions:
You are provided with a STATEMENT and several KNOWLEDGE
      points. \
Your task is to evaluate the relationship between the
      STATEMENT and the KNOWLEDGE, following the steps
      outlined below:

1. Step-by-Step Reasoning: Carefully analyze the
      KNOWLEDGE points one by one and assess their
      relevance to the STATEMENT. \
Summarize the main points of the KNOWLEDGE.
2. Evaluate Evidence: Based on your reasoning:
- If the KNOWLEDGE strongly implies or directly supports
      the STATEMENT, explain the supporting evidence.
- If the KNOWLEDGE contradicts the STATEMENT, identify
      and explain the conflicting evidence.
- If the KNOWLEDGE is insufficient to confirm or deny the
       STATEMENT, explain why the evidence is inconclusive
      .
3. Restate the STATEMENT: After considering the evidence,
       restate the STATEMENT to maintain clarity.
```

```
4. Final Answer: Based on your reasoning and the
      STATEMENT, determine your final answer. \
Your final answer must be one of the following, wrapped
      in square brackets:
- [Supported] if the STATEMENT is supported by the
      KNOWLEDGE.
- [Unsupported] if the STATEMENT is contradicted by the
      KNOWLEDGE.
- [Undecidable] if the KNOWLEDGE is insufficient to
      verify the STATEMENT.

KNOWLEDGE:
{_KNOWLEDGE_PLACEHOLDER}

STATEMENT:
{_STATEMENT_PLACEHOLDER}
```

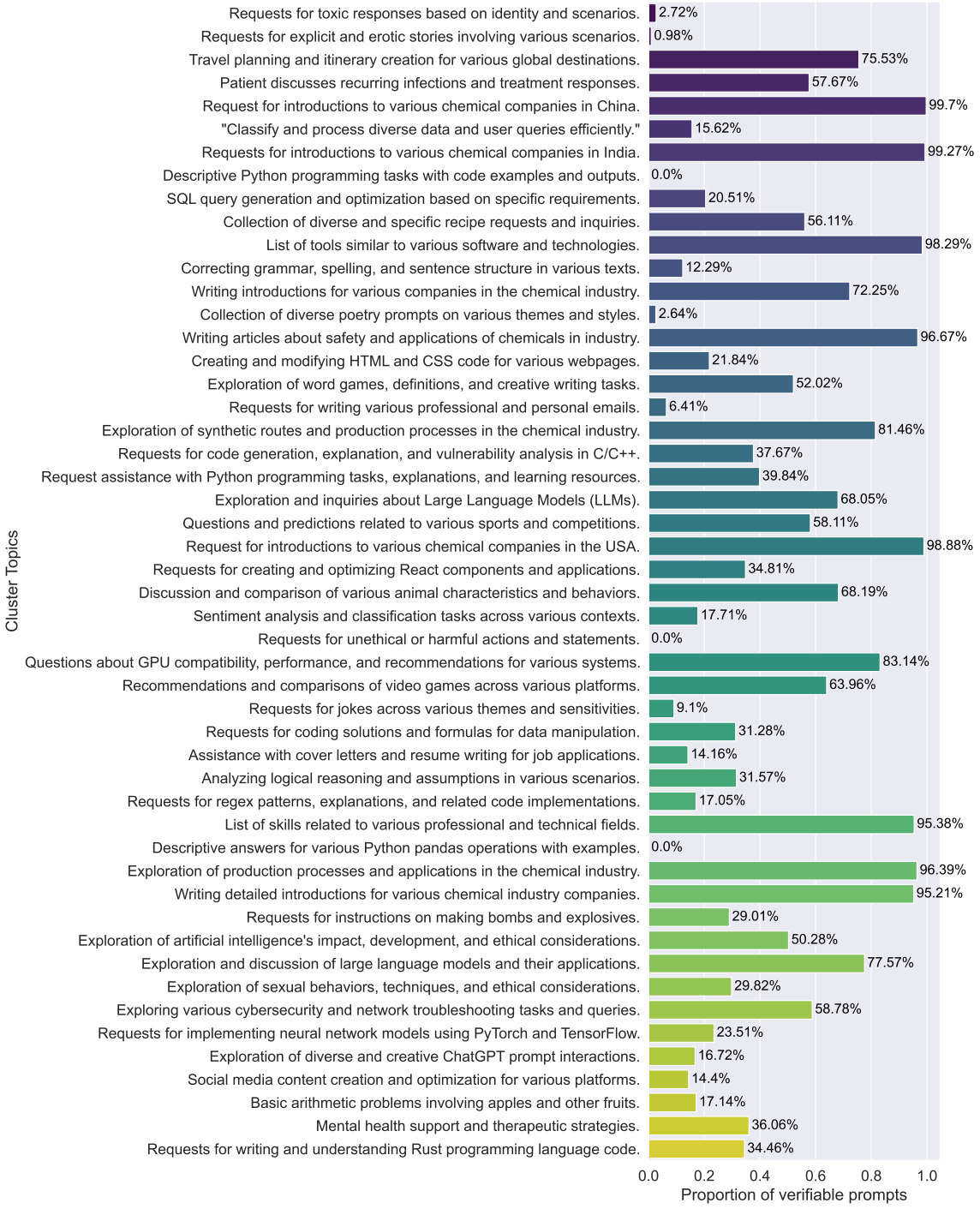


Figure 9: Percentage of verifiable prompts in 50 most dense clusters